

Higher-Order System Response and Dominant Poles

When a third-order system behaves like a second-order one.

Ogata, *Modern Control Engineering*, Ch. 5.

In this tutorial you will:

- approximate a high-order system by its **dominant poles**,
- match DC gain so the approximation is fair, and
- see when the approximation holds and when it breaks down.

Step through with **Ctrl+Enter**, or render a report with **publish**.

A system with three or more poles generally has no simple closed form for its transient specifications. However, if a pair of complex-conjugate poles lies much closer to the $j\omega$ -axis than the remaining poles (a rule of thumb is a real-part ratio of 5:1 or more), that pair **dominates** the transient response, and the system behaves approximately like the second-order system formed by those poles alone – the remaining "fast" poles decay quickly and contribute little after the initial transient.

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Exact Third-Order System

Consider

$$G(s) = \frac{20}{(s^2+2s+5)(s+10)}$$

with poles at $s = -1 \pm j2$ (dominant pair, $\zeta\omega_n = 1$) and $s = -10$ (fast pole, $5\times$ farther from the axis).

```
num = 20;
den = conv([1 2 5],[1 10]);
G_exact = tf(num,den);
poles_exact = pole(G_exact)
```

```
poles_exact =

-10.0000 + 0.0000i
-1.0000 + 2.0000i
-1.0000 - 2.0000i
```

Dominant Second-Order Approximation

Matching DC gain, the second-order approximation built from only the dominant pair is

$$G_{approx}(s) = \frac{K}{s^2+2s+5}, \quad K = G_{exact}(0) \times 5$$

chosen so that $G_{approx}(0) = G_{exact}(0)$.

```

dc_exact = dcgain(G_exact);
K = dc_exact*5;
G_approx = tf(K,[1 2 5]);

fprintf('DC gain exact   = %.4f\n', dc_exact)
fprintf('DC gain approx  = %.4f\n', dcgain(G_approx))

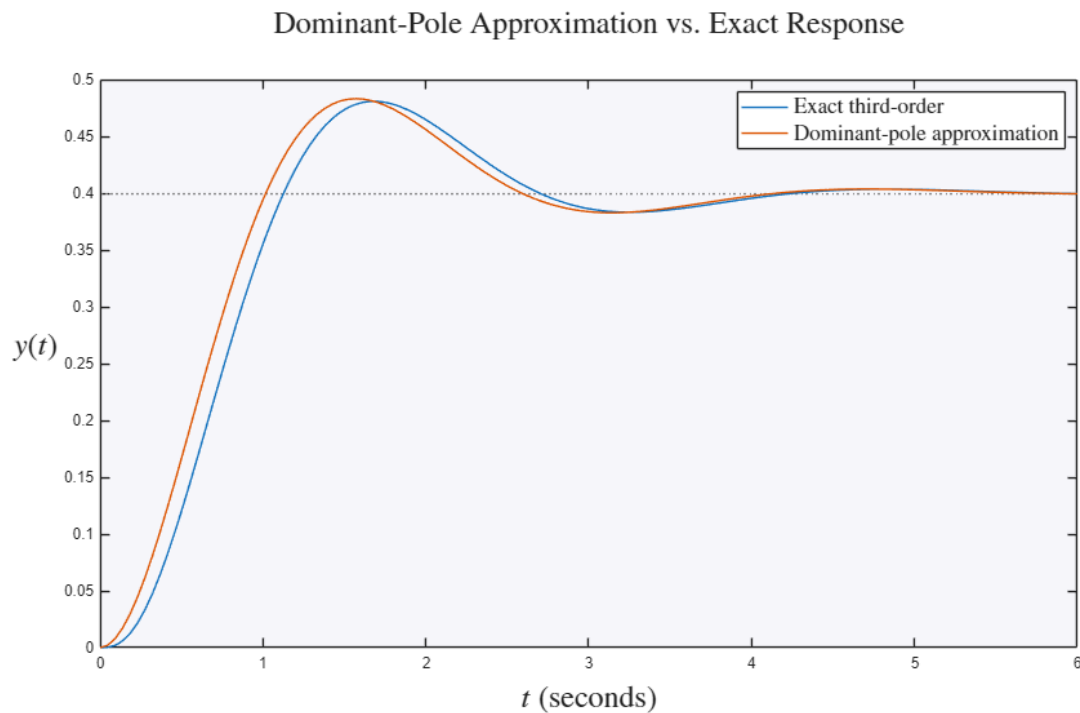
figure
hold on
step(G_exact)
step(G_approx)
hold off
legend('Exact third-order','Dominant-pole approximation','Interpreter','latex','FontSize',14)
title('Dominant-Pole Approximation vs. Exact Response','Interpreter','latex','FontSize',20)
ylabel('$y(t)$','Interpreter','latex','FontSize',20)
set(get(gca, 'YLabel'), 'Rotation', 0)
xlabel('$t$', 'Interpreter','latex','FontSize',20)

```

```

DC gain exact   = 0.4000
DC gain approx  = 0.4000

```



The two curves nearly overlap after the brief initial transient contributed by the fast pole at $s = -10$, confirming the dominant-pole approximation is reasonable here.

When the Approximation Breaks Down

If instead the "extra" pole is comparable in speed to the dominant pair (e.g. at $s = -3$, only $3\times$ the dominant real part), it contributes meaningfully to the transient and the second-order approximation is poor.

```

den_close = conv([1 2 5],[1 3]);
G_close = tf(dc_exact*15, den_close); % rescaled for same DC gain as G_exact

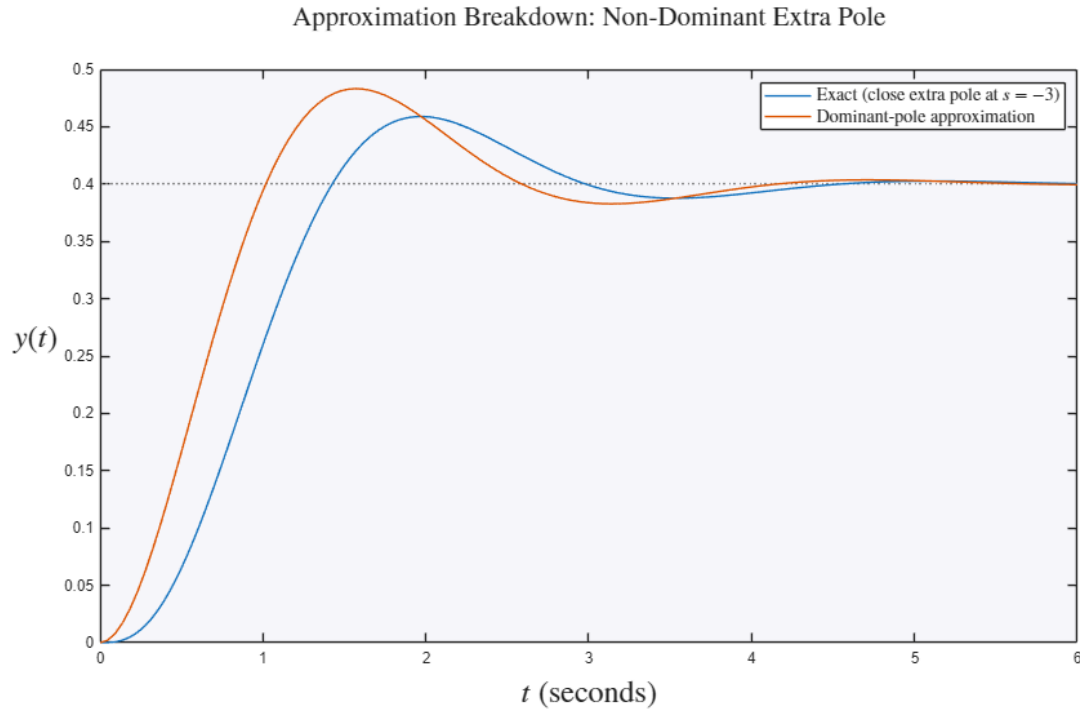
figure
hold on
step(G_close)
step(G_approx)
hold off
legend('Exact (close extra pole at $s=-3$)','Dominant-pole approximation','Interpreter','latex','FontSize',12)

```

```

title('Approximation Breakdown: Non-Dominant Extra Pole','Interpreter','latex','FontSize',18)
ylabel('$y(t)$','Interpreter','latex','FontSize',20)
set(get(gca, 'YLabel'), 'Rotation', 0)
xlabel('$t$', 'Interpreter','latex','FontSize',20)

```



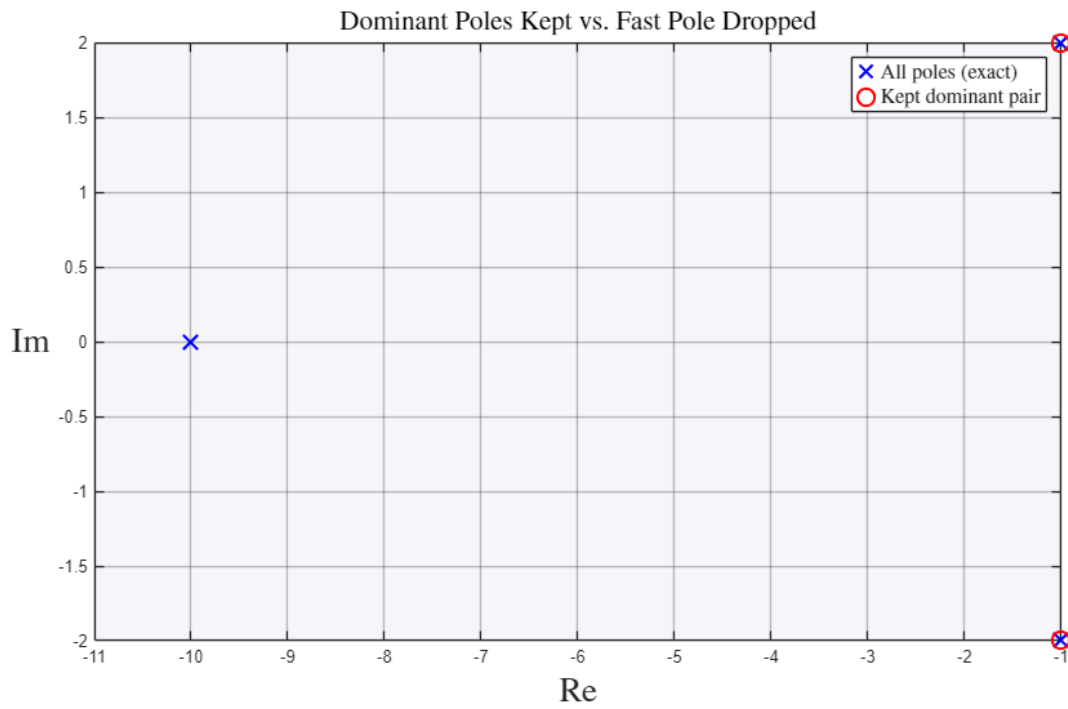
What We Keep vs. What We Drop: The Pole Map

The approximation's logic is visual: keep the slow dominant pair (close to the imaginary axis, sets the bulk of the response) and drop the fast pole far to the left whose mode decays before it matters.

```

figure
plot(real(poles_exact),imag(poles_exact),'bx','MarkerSize',12,'LineWidth',1.5)
hold on
plot(real(roots([1 2 5])),imag(roots([1 2 5])), 'ro','MarkerSize',10,'LineWidth',1.5)
hold off
grid on
legend('All poles (exact)','Kept dominant pair','Interpreter','latex','FontSize',12)
title('Dominant Poles Kept vs. Fast Pole Dropped','Interpreter','latex','FontSize',16)
ylabel('$\mathrm{Im}$','Interpreter','latex','FontSize',20)
set(get(gca, 'YLabel'), 'Rotation', 0)
xlabel('$\mathrm{Re}$','Interpreter','latex','FontSize',20)

```



Try it yourself

- Move the fast pole from -10 to -20 and notice the dominant-pole approximation get even better (the fast mode decays sooner).
- Move it in to -2.5 and watch the approximation break down – it is no longer "fast" relative to the dominant pair.